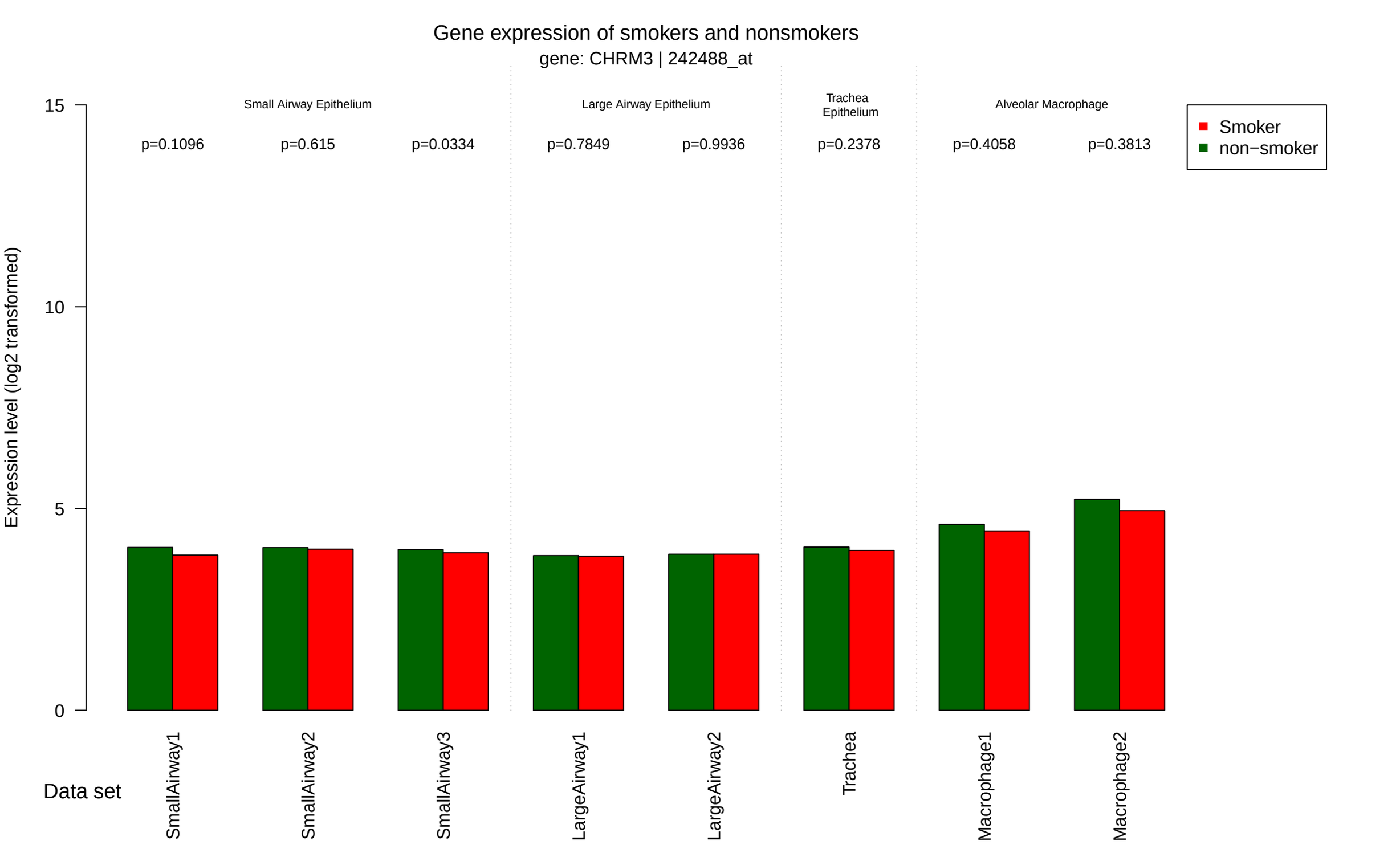
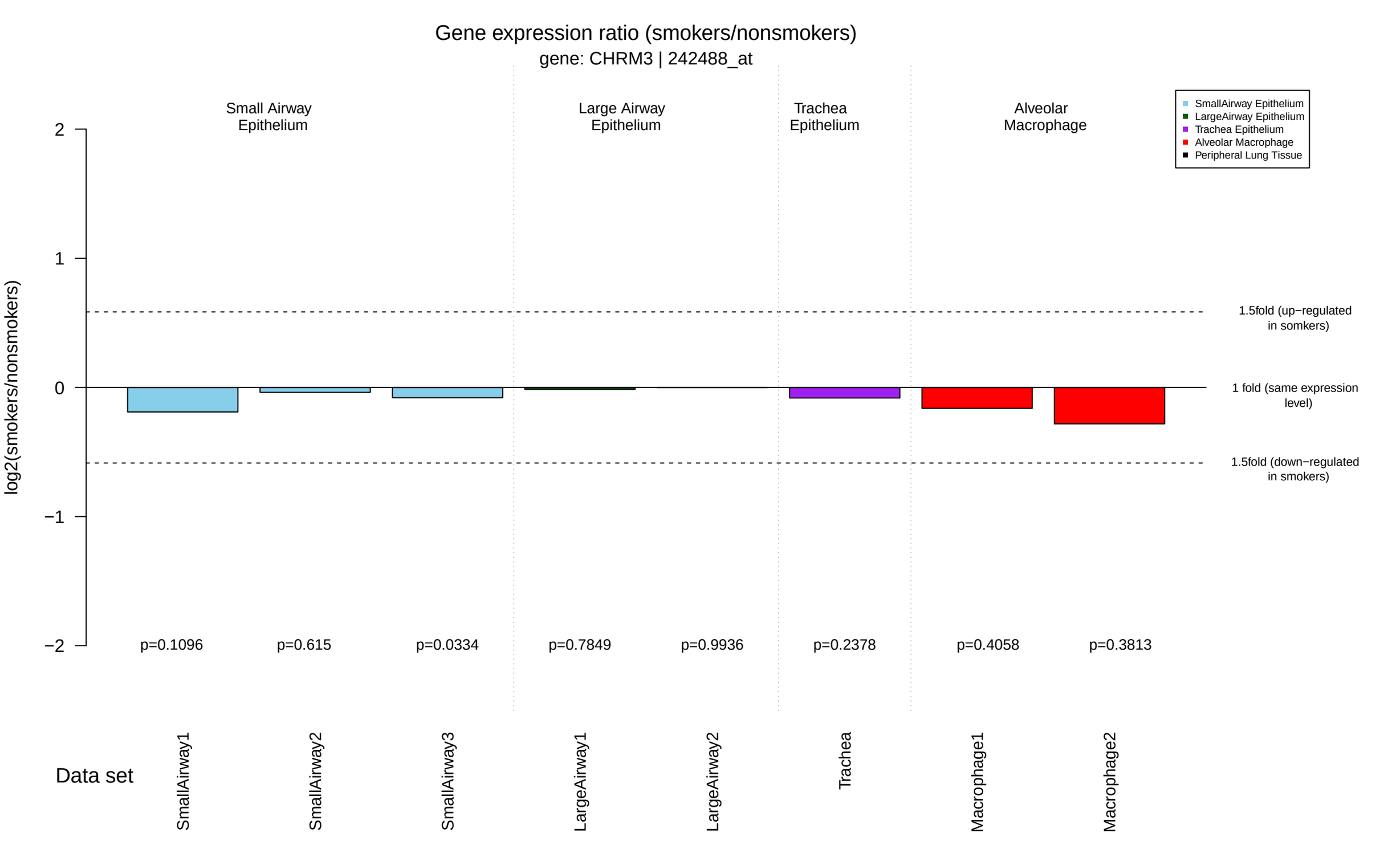
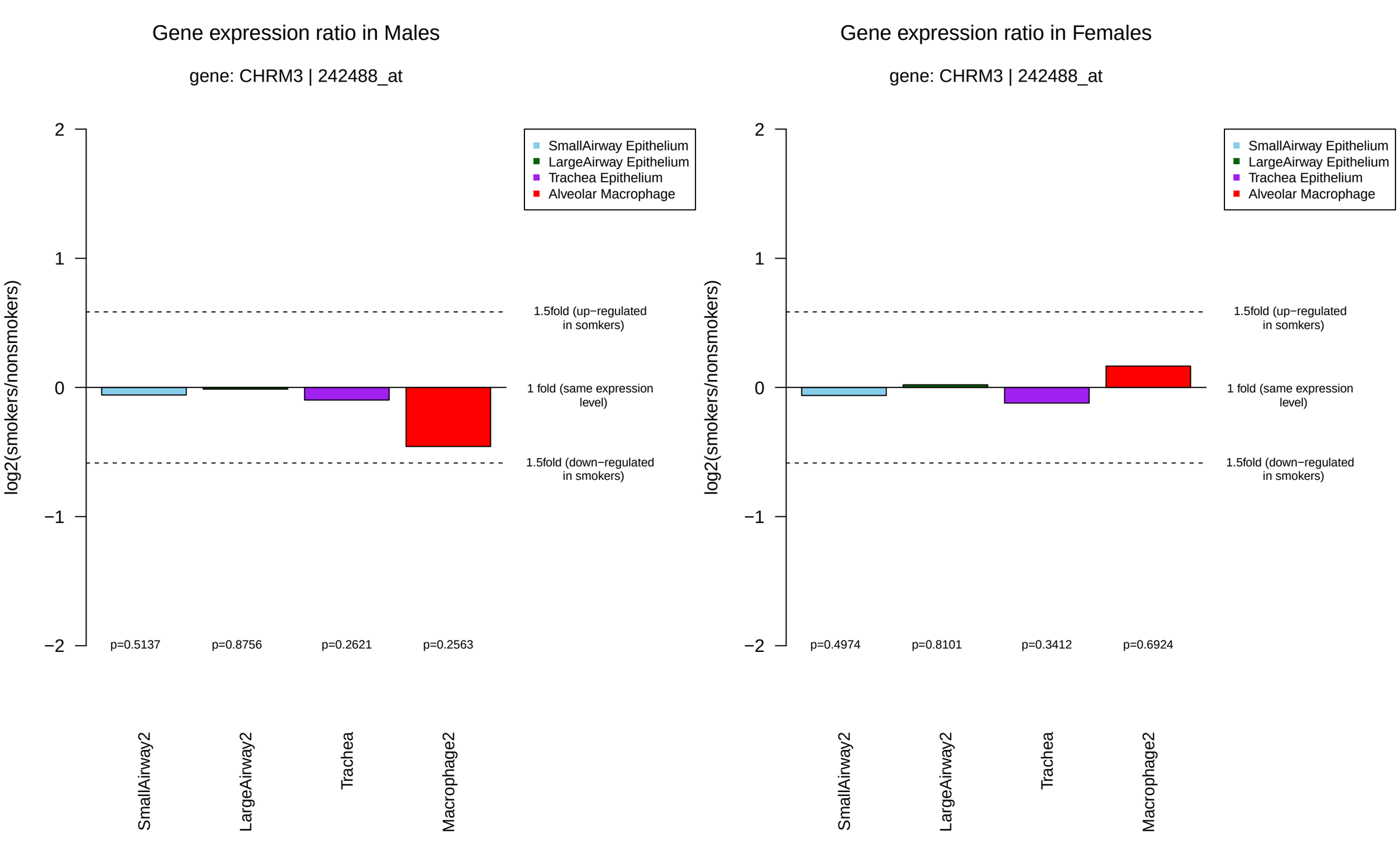
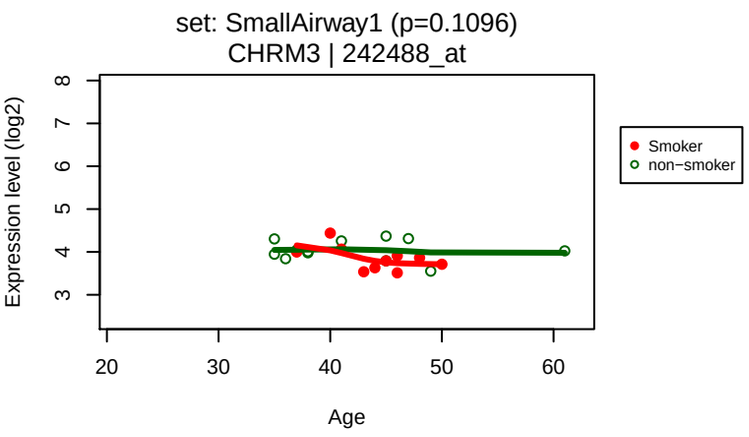
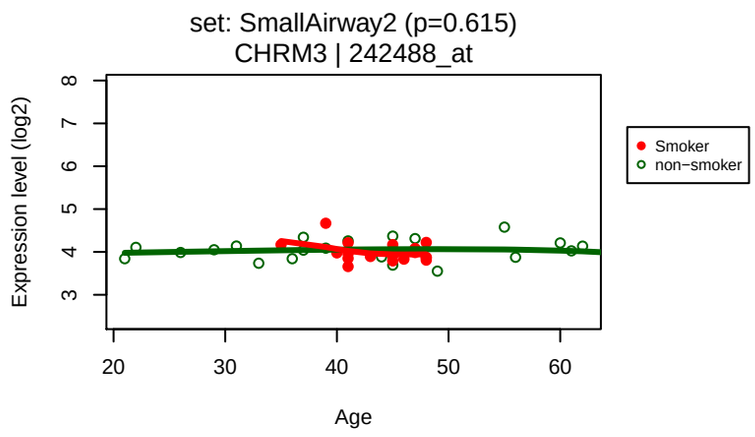
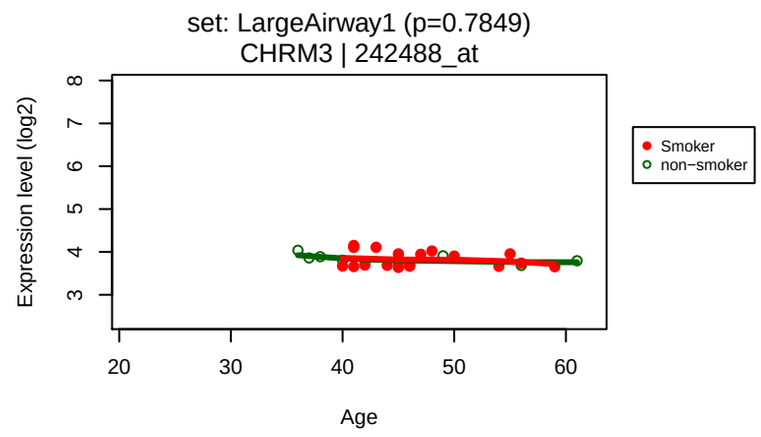

Gene expression of smokers and nonsmokers
gene: CHRM3 | 242488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1096
p=0.615
p=0.0334
p=0.7849
p=0.9936
p=0.2378
p=0.4058
p=0.3813
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

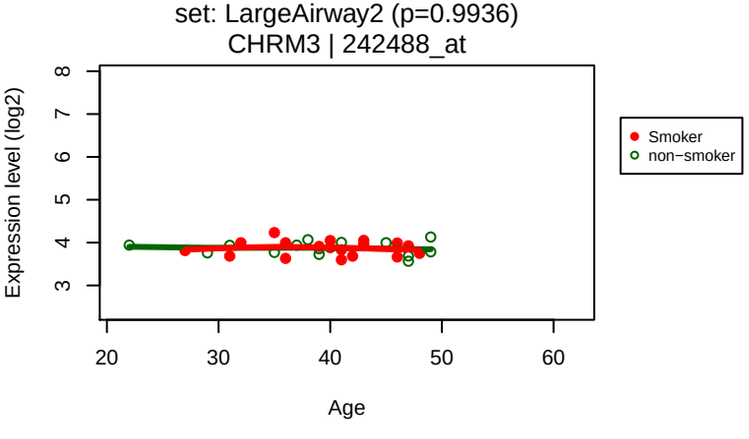

Gene expression ratio (smokers/nonsmokers)
gene: CHRM3 | 242488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1096
p=0.615
p=0.0334
p=0.7849
p=0.9936
p=0.2378
p=0.4058
p=0.3813
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

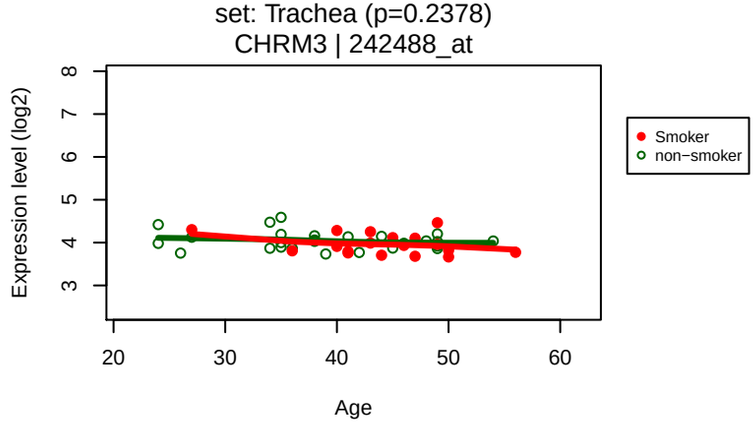

Gene expression ratio in Males
gene: CHRM3 | 242488_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5137
p=0.8756
p=0.2621
p=0.2563
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRM3 | 242488_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4974
p=0.8101
p=0.3412
p=0.6924
SmallAirway2
LargeAirway2
Trachea
Macrophage2

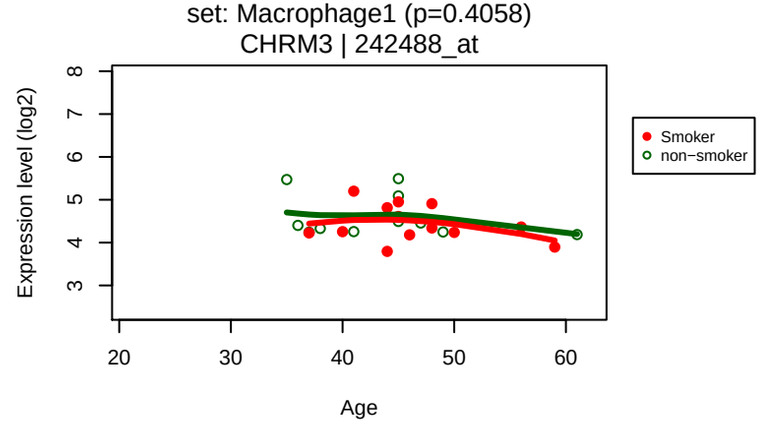

set: SmallAirway1 (p=0.1096)
CHRM3 | 242488_at
Expression level (log2)
Age
Smoker
non−smoker

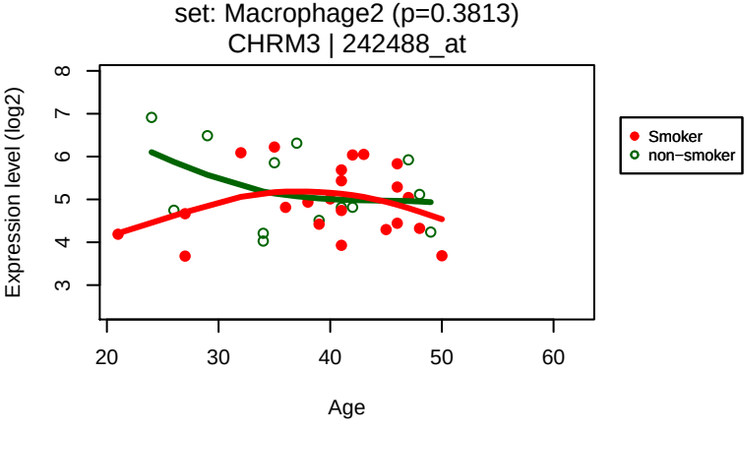

set: SmallAirway2 (p=0.615)
CHRM3 | 242488_at
Expression level (log2)
Age
Smoker
non−smoker

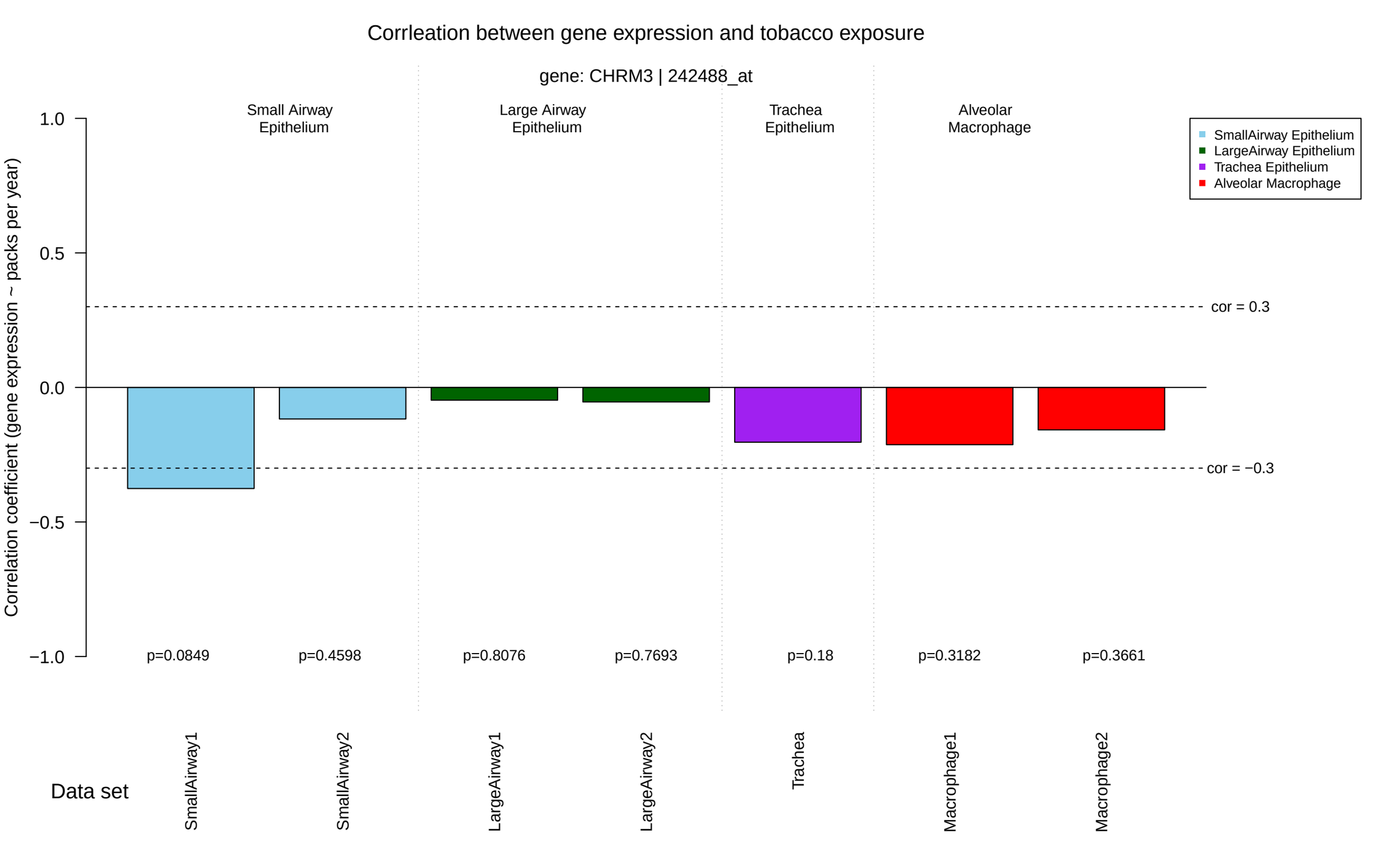

set: LargeAirway1 (p=0.7849)
CHRM3 | 242488_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9936)
CHRM3 | 242488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2378)
CHRM3 | 242488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4058)
CHRM3 | 242488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3813)
CHRM3 | 242488_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRM3 | 242488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0849
p=0.4598
p=0.8076
p=0.7693
p=0.18
p=0.3182
p=0.3661
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2